

9/28/2026

We have proved:

Thm (Unique factorization in $\mathbb{Z}[\sqrt{d}]$, $d = -1, \pm 2, 3$)

Every nonzero, nonunit element of $\mathbb{Z}[\sqrt{d}]$ can be expressed as a product of primes, uniquely up to order and multiplication by units.

This is a nice theorem, but what exactly are the primes and units in $\mathbb{Z}[\sqrt{d}]$?

This question does not have an easy answer for a general d . Nevertheless, our exploration will

produce several satisfactory results and interesting applications.

We start with the case $d < 0$.

Units

Recall that a unit $\varepsilon \in \mathbb{Z}[\sqrt{d}]$ is a (nonzero) element

s.t. $\varepsilon\delta = 1$ for some (nonzero) $\delta \in \mathbb{Z}[\sqrt{d}]$.

We showed

$$\varepsilon \in \mathbb{Z}[\sqrt{d}] \text{ is a unit} \iff |N(\varepsilon)| = 1.$$

If $\varepsilon = x + y\sqrt{d}$, then $N(\varepsilon) = x^2 - dy^2$.

Q 1. What are the units in $\mathbb{Z}[i]$? This corresponds to $d = -1$.

Q 2. What are the units in $\mathbb{Z}[\sqrt{d}]$ when $d < -1$?

Primes in $\mathbb{Z}[i]$

The prime numbers are 2, 3, 5, 7, 11, ...

We know how to check whether a given number is a prime number by looking at its divisors.

But how do we know whether $x + y\sqrt{d}$ is a prime element in $\mathbb{Z}[\sqrt{d}]$? In particular, are prime numbers

also prime elements in $\mathbb{Z}[\sqrt{d}]$?

A sufficient (but not necessary) condition that we have established for a given $\alpha = x + y\sqrt{d}$ to be prime is that

$$|N(\alpha)| = x^2 - dy^2 \text{ is a prime number.}$$

Q3. Can you find the "smallest" prime(s) in $\mathbb{Z}[i]$?

Suppose that $p > 2$ is a prime number. The sufficient condition above says that $\pi = x + iy \in \mathbb{Z}[i]$ is prime in $\mathbb{Z}[i]$ if

$$N(\pi) = \pi \cdot \bar{\pi} = x^2 + y^2 = p.$$

This means p is a sum of two squares!

Q4. What does this imply about p ? In particular, is p prime in $\mathbb{Z}[i]$? And what can we say about $\pi, \bar{\pi}$ and their associates?

Q5. What if p is not a sum of two squares?

Is it prime in $\mathbb{Z}[i]$?

At this point, how can we be sure that we have found all possible primes in $\mathbb{Z}[i]$?

Note that for a prime $\pi \in \mathbb{Z}[i]$,

$$\pi \bar{\pi} = \overbrace{N(\pi)}^{>1} = p_1^{e_1} \cdots p_k^{e_k}.$$

distinct prime numbers

By Euclid's lemma for $\mathbb{Z}[i]$, we have

$$\pi \mid p_i \text{ for some } 1 \leq i \leq k.$$

Moreover, for two distinct prime numbers p_i and p_j ,

$\gcd(p_i, p_j) = 1$. So Bézout's lemma for $\mathbb{Z}$ yields

$$x, y \in \mathbb{Z} \text{ s.t.}$$

$$p_i x + p_j y = 1.$$

Thus $\pi \mid p_i$ and $\pi \mid p_j \Rightarrow \pi \mid 1 \Rightarrow \pi$ is a unit X

Hence, every prime $\pi \in \mathbb{Z}[i]$ divides exactly 1 prime number.

This justifies our search for all primes in $\mathbb{Z}[i]$ by examining divisors of prime numbers.

We will do a full proof on Wednesday that $\pi = x + iy \in \mathbb{Z}[i]$ is prime if and only if one of the following is true :

(1) π is an associate of a prime number $p \equiv 3 \pmod{4}$.

(2) $\pi \mid p$ for some prime number p which is either 2 or $\equiv 1 \pmod{4}$.

When $p = 2$, there are 4 choices for π :

$$\pi = \pm 1 \pm i ;$$

when $p \equiv 1 \pmod{4}$, any choice of π generates 7 other choices!

$$-\pi, \pm \bar{\pi}, \pm i\pi, \pm i\bar{\pi}.$$

These 8 choices are all and only prime divisors of p

in $\mathbb{Z}[i]$. In particular, this implies that there are

exactly 8 ways of writing p as a sum of two squares:

$$p = (\pm x)^2 + (\pm y)^2 = (\pm y)^2 + (\pm x)^2.$$

For example, $10 = (\pm 1)^2 + (\pm 3)^2 = (\pm 3)^2 + (\pm 1)^2$ are all

the representations of 10 as a sum of two squares.

See you on Wednesday!